

AutoPos — 3D Anchor Self-Calibration from Inter-Anchor Ranging Alone

Concept Brief (Paper A: System)

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*Part of a deliberately split two-paper program. This is the **system** paper (Paper A): the working, ground-truth-validated calibration system and the vertical limit it does not cross. A companion **mechanism** paper (Paper B) explains why that vertical limit exists. The two are non-overlapping by design.*

Why this matters. Anchor-based UWB positioning needs accurately known anchor coordinates, and the systems that reach high accuracy obtain them from external infrastructure — surveyed anchor positions or angle-of-arrival base stations (Zihajehzadeh, 2017; Kok, 2015) — exactly the infrastructure missing in ambulatory, out-of-lab settings. Anchor *self*-calibration breaks that circular dependency. Yet even with anchors in hand the field stays well short of the bar it sets itself — the optical “gold standard” the review ties to clinically-relevant decision-making (~ 1 cm): across 37 UWB+IMU systems the *best* reported accuracy is only ~ 5 cm (most 10–80 cm; only $\sim 19\%$ within 10 cm) (Yogesh, 2023, 2024). The shortfall is sharpest on the vertical: only 5 of the 37 attempt 3D at all (the other 32 stay 2D), and even those five take the vertical from IMU fusion over surveyed anchors (Kok, 2015), several adding a biomechanical-model constraint (Zihajehzadeh, 2017). The industrial, aerial side — which this clinical review excludes — does the same: a flying platform resolves height with an onboard laser rangefinder and barometer (Xiang, 2025). Across both directions the vertical is imported from an external aid, *never* recovered from UWB geometry alone. That is the gap AutoPos targets: a body-worn, clinical tag can carry no such aid, so its usable **vertical must come from the UWB geometry itself**.

Why pure-UWB Z, and why 4+4. A body-worn tag does carry one inertial sensor — but no on-body IMU gives a *drift-free, absolute* vertical on its own: an IMU’s height comes from double-integrating acceleration, so it drifts at a rate set by sensor grade — even a premium unit — and is rescued only by gait/biomechanical assumptions the review notes fail for the individual, non-cyclic movement of patients (Yogesh, 2023); a barometer stays coarse and drifts indoors. Recovering Z from the UWB geometry supplies the missing absolute reference and keeps the vertical in the *same* drift-free error regime that already fixes the horizontal, with no second sensor to register or de-drift. The catch is purely geometric: with a single coplanar layer the vertical is *unobservable* in the ideal range-only model — its inter-anchor Cramér–Rao bound is exactly singular, so any apparent height comes from nonidealities or priors, not calibrated geometry, and is far too inaccurate to use. The minimal remedy is the intuitive one — four anchors is the well-known minimum for a 3D fix, so we take that single, horizontally-complete layer and copy it to a second height offset by a Δz , giving a balanced 4+4: the smallest vertical baseline that makes the vertical usable, not an arbitrary denser array nor a coplanar anchor layer plus a Z sensor on the tag. Forcing UWB onto its weakest axis is also what exposed what we name the *delay–layout coupling* (companion paper).

The gap. Existing self-calibration falls into two clusters. One is inter-anchor-only but *2D/coplanar*, or needs a few surveyed reference anchors (De Preter, 2019; Piavanini, 2022; Corbalan, 2023; Ridolfi, 2021). The other reaches *3D* but only by adding an external sensor or mobile platform — LiDAR, SLAM, or visual-inertial odometry (Nguyen, 2025; Yuan, 2024; Delama, 2025) — effectively replacing manual surveying with a robot-assisted survey. The cell “3D + inter-anchor ranging only + joint antenna-delay + no known reference” is empty. It is empty because it is the ill-conditioned regime the field side-steps: De Preter (2019) call 3D self-calibration “very ill-conditioned” and drop to 2D with tape-measured heights; Corbalan (2023) report poor vertical resolution and measure height independently; and commercial RTLS deployments that raise a few anchors to recover the vertical *survey* those raised positions rather than self-calibrate them. Every escape route imports a measured height — the manual step this work removes.

What AutoPos does. On commodity hardware ($8\times$ DWM1001C at two elevation levels) it takes bidirectional SS-TWR inter-anchor ranging through robust fusion (MAD/MVUE) to a joint solve of 3D anchor coordinates *and* per-device antenna delays — no external sensor, no mobile rig, no surveyed reference — and then validates the layout against optical ground truth. The novelty is a *deployment capability and its honest characterization*, not a new estimator: the numerical components (MDS/SDP initialization, Tukey-IRLS robust fitting, alternating delay estimation) are established, and joint anchor+delay estimation is itself prior art (De Preter, 2019; Shah, 2022). (The deployed firmware applies one generic antenna-delay default to every node with no per-device calibration, so the solver’s delays are the residuals that default leaves — the realistic out-of-the-box condition.) What is new is (i) closing the empty 3D inter-anchor-only cell on cheap single-antenna modules; (ii) a *balanced* two-layer 4+4 array

that, *within reference-free self-calibration, and with all eight anchors participating in each fix*, restores the well-conditioned vertical geometry that a single coplanar layer (or a thinned anchor set) lacks — deployed few-anchor raises avoid that wall only by surveying the raised positions, not self-calibrating them (a self-calibration Fisher analysis gives a vertical CRLB exactly singular for a coplanar layer and ~ 63 mm for 4+4 — an RMS lower bound, the same order as the ≈ 62 mm deployed median vertical error). Two ground-truth-free deployment diagnostics also fall out of the program (not headline results): the ranging residual is a misleading quality signal under minimal anchor counts—the self-consistent-but-not-metric effect quantified in Paper B—and estimate dispersion across the dynamic anchor-selection set is a candidate bias indicator a fixed anchor set would hide.

Result, by axis (Std vs absolute kept separate). Repeatability ~ 5 cm 3D / ~ 4 cm vertical; absolute (vs Vicon) ~ 7 cm 3D / ~ 6 cm vertical — UWB-only and self-calibrated. The *vertical* is the point: ≈ 6 cm median absolute from pure UWB, on an axis 32 of 37 reviewed systems avoid entirely and that the remaining few reach only by importing an external height reference. The defensible claim is a usable vertical recovered from UWB alone under the body-worn constraint that rules out external height aids — not the smallest absolute number (the best fusion-plus-surveyed system still edges the 3D total).

Anchor visibility. At every position 92.2% of frames ranged to all eight anchors and 99.9% to at least seven (per-position median 8/8; link-level success 99.0%). A seven-of-eight fix is still non-coplanar and over-determined, so the vertical stays well-conditioned in 99.9% of frames and the reported medians are effectively whole-array fixes.

Honest limits. The preliminary ~ 5 cm numbers are *repeatability*, not accuracy — only the optical-ground-truth numbers are reported as accuracy. The 4+4 geometry is *necessary but not sufficient* for the vertical: it completes the geometric observability, but a systematic vertical bias remains — a *tag-side delay* aliasing into the vertical, not a geometry deficit — which the companion mechanism paper (Paper B) analyzes as the tag-side face of its delay-layout coupling result. These numbers also presuppose that every fix draws on all eight anchors of the two-layer array — it is the anchor participation, not the ranging protocol (sequential and broadcast are identical for a static fix), that conditions the vertical: a fix from fewer anchors (e.g. a four-anchor subset) hits a much larger vertical CRLB, and a coplanar configuration returns the vertical to the singular regime by construction; neither reproduces these numbers. All numbers are from a single room with 24 static tag positions; multi-room transfer is unproven. The headline 3D median (≈ 73 mm) also benefits, in part, from the production layout’s expanded scale compensating an uncorrected tag-side delay — the companion paper’s mechanism; with that delay left at zero the metric-correct layout gives a larger median. The absolute vertical is also referenced to the optical marker, so a fixed marker-to-antenna-phase-centre offset would shift the absolute value (Paper B’s phase-center sensitivity check finds the layout ranking robust to marker offsets beyond 10 mm).

Scope. To our knowledge this is the first demonstration that the *vertical* axis of a UWB anchor array can be self-calibrated to the centimetre level (≈ 62 mm median absolute vs Vicon, against a Fisher-predicted ~ 63 mm inter-anchor vertical CRLB for 4+4) from inter-anchor ranging alone on commodity single-antenna modules — a reference point that later reference-free pure-UWB vertical self-calibration in this regime can be measured against. The full enabling combination is stated once (3D self-calibration; inter-anchor ranging only; joint per-device antenna-delay estimation; commodity DWM1001C modules; balanced 4+4 array with all eight anchors participating in each fix; optical-ground-truth validation — no survey, no external Z sensor, no mobile platform), and ships the CRLB script and ranging data so the floor is independently reproducible.

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